DNA Applications in Nanobiotechnology

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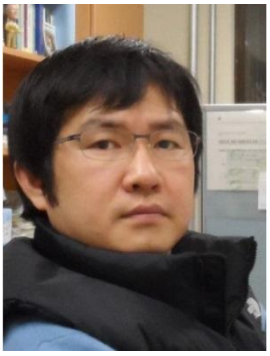
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Abstract

Nanobiotechnology has evolved into a unique interdisciplinary field involving physics, materials science, chemistry, biology, computer science, and multiple engineering fields. Likewise, DNA nanotechnology is a quickly developing field with essentially no overwhelming technical difficulties inhibiting progress toward designing and fabricating new shapes of DNA nanostructures in all dimensions. In this field, researchers create artificial DNA sequences to self-assemble into target molecular nanostructures. The well understood Watson–Crick base-pairing rules are used to encode assembly instruction directly into the DNA molecules which provide basic building blocks for constructing functionalized nanostructures with two major features: self-assembly and self-alignment. In this talk, we present on various DNA nanostructures with specific shapes and patterns generated by implementation of certain rules. In addition, we address specific applications using DNA which show feasibility to construct various physical devices and biological sensors made of DNA molecules.

Biography

Sung Ha Park is an associate professor of Physics, Sungkyunkwan Advanced Institute of NanoTechnology (SAINT), and Biomedical Engineering (BME) at Sungkyunkwan University, Korea. He got his Ph.D. in physics at Duke University, USA (2005). From 2005 to 2007, he worked at Caltech, USA as a postdoctoral researcher studying the subject of DNA Nanotechnology and DNA computing. He is interested in researches mainly on physical applications of DNA nanotechnology, DNA algorithmic assembly, bionanotechnology, and mathematical biology. Until present, he have published more than 100 research articles in scholarly journals, such as Science, Nature Nanotechnology, Nano Letters, ACS Nano, JACS and Angewandte Chemie. Currently, he is a member of the Korean Physical Society and the Korean Biophysical Society and serves as an associate editor for a journal of NANO, World Scientific.



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